E^2UADE: A Principled Approach to a Consensus-Driven Evaluation Framework for Unsupervised Anomaly Detection

Buddhi Sagar Poudel[1*] and Arpan Rai[1]

[1]Department of Computer Science and Engineering, Nepal Engineering College, Affiliated to Pokhara University, Pokhara, Nepal.

*Corresponding author(s). E-mail(s): iamsagar099@gmail.com;
Contributing authors: mail.arpanrai@gmail.com;

Abstract

Evaluating unsupervised anomaly detection models is fundamentally challenging due to the pervasive absence of ground truth labels in real-world applications. Existing evaluation paradigms often rely on unrealistic assumptions, suffer from poor comparability, or fail to provide interpretable and stable assessments, thereby hindering scientific and practical progress. This paper introduces E^2UADE (Ensemble-based Unsupervised Anomaly Detection Evaluation), a principled framework designed to overcome these limitations. E^2UADE leverages the collective signal from a diverse ensemble of unsupervised detectors to evaluate individual model performance across three interpretable dimensions: consensus (Intersection Ratio), diversity (Diversity Factor), and ranking fidelity (Normalized Kendall's Tau). The framework incorporates robust median-based ensemble aggregation, a rigorously defined fixed-K thresholding strategy ensuring fair comparability, explicit handling of unknown contamination through parameterization and sensitivity analysis, and a quantitative bootstrap-based assessment of evaluation stability. By deliberately avoiding distorting normalization schemes and emphasizing methodological rigor, E^2UADE provides a theoretically sound, practically applicable, and interpretable methodology for reliably comparing and selecting unsupervised anomaly detection models in fully unlabeled settings.

Keywords: Anomaly detection, ensemble methods, model evaluation, model selection, Kendall's tau, unsupervised learning.

1 Introduction

Anomaly detection - the identification of rare and atypical patterns that deviate from expected norms, plays a critical role in diverse fields such as cybersecurity, finance, industrial monitoring and healthcare [2], [8]. In principle, supervised anomaly detection models offer powerful solutions when labeled data for both normal and anomalous behaviors are available. However, in most real-world settings, anomalies are inherently rare, evolve dynamically, and are costly or impractical to label, leading to a widespread scarcity of ground truth [20]. This makes supervised evaluation unreliable or infeasible, necessitating a reliance on unsupervised anomaly detection methods that infer anomalousness solely from the intrinsic structure of unlabeled data.

However, evaluating unsupervised anomaly detectors in the absence of labels presents a profound challenge: Without ground truth, it becomes difficult to determine which model performs better, to tune hyperparameters, or to build trust in unsupervised systems [11]. This challenge manifests itself in several critical ways that have not been adequately addressed in the existing literature.

First, practitioners face a fundamental model selection dilemma in unsupervised settings: When multiple competing models (e.g. Isolation Forest, Autoencoders, LOF) are applied to the same dataset, they frequently produce substantially different anomaly sets [1], [6]. For example, if an Autoencoder identifies the anomaly set Y_{ae} with m anomalies and Isolation Forest identifies set Y_{if} with m anomalies, the differences of the set $Y_{ae} - Y_{if}$ and $Y_{if} - Y_{ae}$ are rarely empty [18]. This divergence occurs because each model employs a distinct underlying mechanism to identify anomalousness. Autoencoders flag points with high reconstruction error, while Isolation Forest leverages path length normalization [41], [31]. In production environments where the deployment of multiple models is often impractical due to computational, maintenance, or interpretability constraints, how does one objectively select a single "best" model without labeled validation data [43]?

Second, this model selection problem is magnified when addressing novel, complex tasks that lack prior domain knowledge or established benchmarks [47]. In such scenarios, domain experts may have incomplete understanding of what constitutes an anomaly, labeled examples are unavailable, and the nature of anomalies may shift over time due to concept drift [34]. The absence of a reference "ground truth" creates a circular dependency: we need effective models to identify anomalies, but we also need knowledge of anomalies to validate model effectiveness [23].

Third, the heterogeneity of the evaluation metrics between model types further complicates comparative evaluation [18], [38]. Different anomaly detection paradigms not only produce different anomaly sets but also use fundamentally incomparable scoring mechanisms - Autoencoders generate reconstruction errors (typically MSE-based), Isolation Forest employs path length distributions, and Local Outlier Factor uses local density deviations [33]. These diverse metrics have different scales, distributions, and interpretations, making direct comparison problematic. Furthermore, naive score normalization approaches often introduce distortions that compromise evaluation integrity [6], [18].

Early approaches have attempted to assess intrinsic model properties such as clustering validity indices or stability under perturbations, but these often correlate poorly with real-world anomaly detection performance [6]. Alternative strategies like pseudo-labeling [5], self-supervised tasks [21], or synthetic anomaly injection [23] offer partial signals but suffer from bias amplification, questionable task relevance, or unrealistic anomaly generation [14].

A promising direction is ensemble-based evaluation, where agreement among a diverse set of detectors serves as a proxy for ground truth [1], [29], [40]. However, prior ensemble approaches suffer from critical methodological weaknesses: inconsistent comparability due to varying thresholds, sensitivity to aggregation biases, inadequate handling of the unknown contamination factor, and a lack of stability analysis.

To address these limitations, this paper introduces E^2UADE (Ensemble-based Unsupervised Anomaly Detection Evaluation)—a robust, interpretable, and methodologically principled framework for evaluating unsupervised anomaly detection models without any reliance on labels. E^2UADE leverages robust ensemble aggregation, a fixed-K thresholding strategy to ensure comparability, explicit parameterization of the contamination assumption, and a bootstrap-based stability assessment. By systematically addressing model selection, contamination uncertainty, and evaluation stability challenges, E^2UADE provides a reliable basis for comparing and selecting unsupervised detectors in fully unlabeled environments.

2 Related Work: Evaluating Unsupervised Anomaly Detection

The challenge of evaluating unsupervised anomaly detectors without ground truth has spurred various approaches, often inheriting strengths and weaknesses from broader unsupervised learning evaluation practices. We categorize and synthesize these approaches below, highlighting limitations that necessitate the E^2UADE framework.

2.1 Intrinsic Model Properties and Stability

One group of methods focuses on internal characteristics of models or their outputs. This includes assessing the quality of density estimates or cluster separation (drawing from clustering validation indices like Silhouette Coefficient [4]), evaluating the reconstruction error for reconstruction-based models like Autoencoders [3], or measuring the stability of outlier scores/rankings under data perturbations like subsampling [26].

The underlying assumption is that well-behaved models are better anomaly detectors. However, a primary limitation is often weak correlation between these internal/stability properties and the actual ability to correctly identify diverse, real-world anomalies [11]. Furthermore, comparing such metrics across fundamentally different model architectures is often problematic. For example, comparing an Autoencoder's reconstruction error with Isolation Forest's path length or LOF's local density deviation provides little insight into relative effectiveness, as these metrics operate in fundamentally different spaces with incomparable units and distributions [6], [18].

Boukerche et al. [32] demonstrated that models optimized for certain intrinsic properties often fail to generalize to real anomalies, while Goix et al. [37] showed that stability-based metrics frequently favor overly conservative models that miss subtle but important anomalies. This disconnect between intrinsic properties and actual anomaly detection performance severely limits the practical utility of such approaches for model selection in production environments [43].

2.2 Auxiliary Tasks and Data

Another category attempts to create surrogate evaluation signals. Pseudo-labeling techniques use high-confidence predictions from an initial model run as temporary labels [5], while self-supervised approaches might leverage pretext tasks assuming performance correlates with anomaly detection capability [21]. Synthetic anomaly injection modifies the dataset by adding artificial outliers, allowing the use of standard supervised metrics [23].

While innovative, these methods face significant hurdles: pseudo-labeling can amplify initial model biases; self-supervised task relevance to anomaly detection is not guaranteed; and the validity of synthetic injection hinges entirely on the difficult-to-achieve realism and representativeness of the generated anomalies [14].

Bovenzi et al. [31] empirically demonstrated that pseudo-labels tend to propagate and amplify the biases of the initial model, creating a self-reinforcing evaluation system that may systematically favor certain anomaly types while ignoring others. The relationship between self-supervised learning tasks and anomaly detection is often tenuous; [44] showed that models performing well on rotation prediction or contrastive learning objectives can still perform poorly on real anomaly detection tasks when the underlying anomaly mechanisms differ from those captured by the pretext task.

The synthetic anomaly generation approach, while conceptually appealing, struggles with the fundamental challenge of creating realistic anomalies without prior knowledge of what constitutes anomalousness in the specific domain [46], [23]. As shown by Campos et al. [34], synthetic anomalies often lack the complex, multidimensional patterns of real anomalies, leading to overly optimistic performance estimates and models that detect artificial patterns rather than genuine anomalies.

2.3 Ensemble-Based Evaluation

Given that ensembles often improve anomaly detection robustness [1], using ensemble outputs as a reference for evaluating individual models is a promising direction [29]. The core idea is that consensus among diverse detectors provides a more reliable signal than any single model. Models can be compared based on their agreement with the ensemble output. However, practical implementations frequently suffer from critical methodological weaknesses:

2.3.1 Comparability Issue

Often, each model M_k determines its own anomaly set A_k, leading to varying sizes $|A_k|$. Comparing metrics like intersection ratio then involves comparing proportions of different denominators, undermining fair comparison [6]. Goswami et al. [38] demonstrated that even minor variations in the threshold selection strategy can lead to dramatic shifts in relative model rankings, rendering evaluation results arbitrary and unstable. This issue becomes particularly acute when comparing models with fundamentally different score distributions, such as Isolation Forests and deep autoencoder-based methods [43].

2.3.2 Aggregation Sensitivity

Using the mean score for the ensemble reference is simple but highly sensitive to models with disparate score scales or outlier behavior, potentially skewing the reference [18]. Bovenzi et al. [31] presented empirical evidence showing that arithmetic mean aggregation can be dominated by a single model with extreme scores, effectively reducing a diverse ensemble to a single-model evaluation. This vulnerability undermines the core rationale for ensemble-based evaluation—harnessing diverse perspectives to approximate ground truth [30].

2.3.3 Contamination Factor Uncertainty

The unknown true proportion of anomalies (p, sometimes denoted as κ) is rarely handled robustly. It might be ignored, crudely estimated [16], or addressed via post-hoc normalization based on detected counts ($|A_k|$), which can severely distort scores. Emmott et al. [36] demonstrated that even small errors in contamination rate estimation can substantially alter evaluation outcomes. Steinbuss and Böhm [23] further showed that different models perform optimally under different contamination assumptions, making this parameter crucial for fair comparison.

2.3.4 Stability Neglect

The evaluation score depends on the specific ensemble M. The stability or reliability of this evaluation concerning variations in M is typically not quantified [29]. Aggarwal and Sathe [30] highlighted that ensemble evaluation results can vary significantly based on ensemble composition, yet this uncertainty is rarely reported or incorporated into model selection decisions. This neglect of stability assessment leaves practitioners without confidence measures for the evaluation itself.

This review reveals a clear need for an unsupervised evaluation framework, particularly within the promising ensemble-based paradigm, that systematically addresses these limitations. E^{2}UADE is explicitly designed to fill this methodological gap.

3 The E^{2}UADE Framework: Rationale and Components

E^{2}UADE builds upon the ensemble evaluation paradigm but incorporates specific design choices justified by the need for robustness, comparability, and interpretability, addressing the limitations identified in Section 2.

3.1 Ensemble Reference Signal

E^{2}UADE uses a diverse ensemble $M = \{M_1, ..., M_m\}$ as its reference. The rationale is the "wisdom of crowds" principle: a consensus signal derived from multiple, diverse detection approaches is likely more reliable than any single detector, serving as a proxy for ground truth.

Robust Aggregation (Median): To generate the ensemble score $s_{ensemble}$ for each instance, E^{2}UADE uses the median of individual model scores s'_k (standardized as per Subsection 4.3.1). The median is statistically robust to outlier values, unlike the mean [12], [15], minimizing the influence of models with aberrant score distributions and addressing the aggregation sensitivity limitation (see Subsubsection 2.3.2).

3.2 Consistent Evaluation Scope (Fixed-K Thresholding)

A core innovation of E^{2}UADE is its thresholding strategy. Instead of allowing each model to determine its own number of anomalies, E^{2}UADE operates based on a predefined target anomaly count, K.

Procedure: Both individual models (yielding anomaly set A_k) and the ensemble (yielding $A_{ensemble}$) are evaluated based on identifying the top K instances with the highest anomaly scores. By ensuring $|A_k| = |A_{ensemble}| = K$, E^{2}UADE establishes a level playing field. Metrics comparing these sets (like Intersection Ratio) are calculated as proportions of the same quantity (K), making direct comparison across models meaningful and addressing the critical comparability issue (see Subsubsection 2.3.1) highlighted by [6].

3.3 Explicit Contamination Handling

E^{2}UADE directly confronts the uncertainty surrounding the true contamination factor $p = K/n$.

Parameterization: The number of evaluated anomalies K (and thus the implied contamination rate $p = K/n$) is an explicit input parameter to the framework.

Transparency: The framework reports the operational $p = K/n$ alongside the performance metrics, making the evaluation context clear.

Sensitivity Analysis: Users are strongly encouraged to run E^{2}UADE across a range of plausible K values to understand how model performance varies with this assumption (see Subsection 6.3). This approach avoids the score distortions associated with post-hoc normalization based on variable detected counts. It makes the evaluation explicitly conditional on the assumed p, which is scientifically more transparent than implicitly absorbing this uncertainty into a potentially flawed normalization [16].

3.4 Core Evaluation Dimensions & Metrics

Within this structured setup (median aggregation, fixed-K), E^{2}UADE assesses model performance relative to the ensemble across key interpretable dimensions.

Consensus vs. Diversity: How much does a model agree with the ensemble's top K findings (measured by Intersection Ratio, IR), and how much does it contribute unique findings within that top K (measured by Diversity Factor, DF)?

Ranking Fidelity: How well does the model's overall ranking of all data points align with the ensemble's ranking (measured by Normalized Kendall's Tau, NKT)?

These dimensions provide a multi-faceted view of model behavior. The specific metrics are explained in Section IV.

4 E^2UADE Algorithm Specification

This section formally defines the E^2UADE algorithm, detailing its inputs, outputs, procedural steps, and the significance of its computed metrics.

4.1 Inputs

- **D**: Dataset of n instances $\{x_1, \ldots, x_n\}$.
- **M**: Ensemble of m diverse unsupervised AD models $\{M_1, \ldots, M_m\}$.
- **K**: Target number of top anomalies to evaluate (integer, $1 \leq K < n$).
- **a, b, d**: Weights for WDCRS (non-negative, representing relative importance of Consensus (IR), Diversity (DF), and Ranking Fidelity (NKT) respectively; often normalized to sum to 1).

4.2 Outputs (for each model M_k)

- **IR(M_k)**: Intersection Ratio score.
- **DF(M_k)**: Diversity Factor score.
- **NKT(M_k)**: Normalized Kendall's Tau score.
- **WDCRS(M_k)**: Weighted composite score.
- **p**: Assumed contamination rate (K/n).
- $WDCRS_{\mathbf{std.dev}}(M_k)$: Evaluation stability score (computed in Section V).

4.3 Procedure

4.3.1 Score Generation & Standardization

a) For each $M_k \in M$, compute raw anomaly scores $s_k(x_i)$ for all $x_i \in D$.

b) (Recommended unless scores are known to be comparable): Apply robust score standardization. We suggest using Median Absolute Deviation (MAD)-based standardization:

$$s'_k(x_i) = \frac{s_k(x_i) - \text{median}(s_k)}{\text{MAD}(s_k) \cdot c + \epsilon} \tag{1}$$

where $\text{MAD}(s_k) = \text{median}(|s_k - \text{median}(s_k)|)$, c is a consistency constant (≈ 1.4826) [22], and ϵ is a small constant (e.g., 10^{-8}). Let s'_k denote the scores used henceforth.

4.3.2 Ensemble Score Calculation

For each $x_i \in D$, compute $s_{ensemble}(x_i) = \text{median}(\{s'_1(x_i), \ldots, s'_m(x_i)\})$.

4.3.3 Fixed-K Anomaly Set Identification

a) For each model M_k, find A_k = set of K instances with the highest scores s'_k. Ensure $|A_k| = K$. Use a consistent tie-breaking method.

b) Find $A_{ensemble}$ = set of K instances with the highest scores $s_{ensemble}$. Ensure $|A_{ensemble}| = K$, using the same tie-breaking method.

4.3.4 Metric Calculation

- **Intersection Ratio (IR):** Measures the proportion of a model's top K anomalies that are also top K by the ensemble. High IR indicates strong agreement.

$$\mathrm{IR}(M_k) = \frac{|A_k \cap A_{ensemble}|}{K} \tag{2}$$

- **Diversity Factor (DF):** Measures the proportion of a model's top K anomalies that are unique compared to the ensemble's top K. High DF suggests the model finds potential anomalies missed by the consensus.

$$\mathrm{DF}(M_k) = \frac{|A_k \setminus A_{ensemble}|}{K} = 1 - \mathrm{IR}(M_k) \tag{3}$$

- **Normalized Kendall's Tau (NKT):** Evaluates the consistency of the overall ranking of all n instances between the model and the ensemble, using Kendall's rank correlation coefficient (τ) [17].

$$\mathrm{NKT}(M_k) = \frac{\tau(s'_k, s_{ensemble}) + 1}{2} \tag{4}$$

- **Contamination Level (CL):** Explicitly reports the assumed contamination rate $p = K/n$.
- **Weighted Composite Score (WDCRS):** Provides a single summary score reflecting user-defined priorities.

$$\mathrm{WDCRS}(M_k) = a \cdot \mathrm{IR}(M_k) + b \cdot \mathrm{DF}(M_k) + d \cdot \mathrm{NKT}(M_k) \tag{5}$$

This core procedure yields the primary evaluation outputs. Assessing the stability of these outputs is addressed next.

5 Evaluation Stability Assessment

The E^{2}UADE results depend on the chosen ensemble M. To assess their reliability, we employ a bootstrap procedure [10].

5.1 Motivation

A stable evaluation framework should produce consistent scores for a given model M_k even if the reference ensemble is slightly perturbed. Quantifying this stability provides confidence in the robustness of the computed scores.

5.2 Bootstrapping Procedure

Let r be the number of bootstrap iterations (e.g., $r = 100$).

1. Initialize storage: `bootstrap_scores` = {k: [] for k in range(m)}.

2. For iteration $b = 1$ to r:
   (a) Create bootstrap ensemble M_{boot} by sampling m models from M with replacement.
   (b) Calculate bootstrap ensemble score s_{boot} using median aggregation on scores from models in M_{boot}.
   (c) Determine bootstrap anomaly set A_{boot} (Top K based on s_{boot}).
   (d) For each original model M_k (using its original scores s'_k):
      (i) Recalculate IR_{boot}, DF_{boot}, and NKT_{boot} relative to A_k and A_{boot}.
      (ii) Calculate WDCRS_{boot} using these perturbed metrics.
      (iii) Append WDCRS_{boot} to `bootstrap_scores[k]`.
3. For each model M_k: Compute the stability metric:

$$WDCRS_{\text{std.dev}}(M_k) = \text{std.dev}(\texttt{bootstrap\_scores}[k]) \quad (6)$$

5.3 Interpretation and Output

A low $WDCRS_{\text{std.dev}}(M_k)$ indicates the model's score is stable and reliable. A high value suggests sensitivity to ensemble composition, warranting caution. While no fixed threshold is prescribed, values < 0.1 or 0.15 could be considered sufficiently stable, depending on the application.

6 Practical Implementation and Validation

This section details practical aspects of applying E^2UADE and validating its utility.

6.1 Implementation Workflow

Implementing E^2UADE involves standard data science workflows using libraries like Scikit-learn [19], PyOD [27], NumPy [13], and SciPy [25]. The bootstrapping loop can be parallelized.

6.2 Computational Complexity

Runtime is influenced by base model complexity $O(\text{models})$, dataset size $O(n)$, ensemble size $O(m)$, and bootstrap iterations $O(r)$. Key steps include model execution $O(\text{models})$, standardization/aggregation/thresholding $O(m \cdot n \log n)$, and bootstrapping $O(r \cdot m \cdot n \log n)$.

6.3 Parameter Selection Guidance

- **Ensemble M:** Requires a diverse set of quality models (distance, density, isolation, reconstruction, etc.). Size $m \geq 5$ is recommended for stability.
- **Target anomaly count K:** Base on domain expertise, operational limits, or a sensitivity analysis. Running E^2UADE for multiple K values (e.g., contamination rates $p \in \{0.01, 0.03, 0.05, 0.1\}$) is essential to assess ranking robustness.
- **WDCRS Weights (a, b, d):** Define evaluation goals. For example:
  - *Balanced:* a=0.5, b=0.2, d=0.3

– *Consensus-Focused:* a=0.7, b=0.1, d=0.2
– *Novelty-Seeking (Use Cautiously):* a=0.3, b=0.4, d=0.3

Always report individual metrics (IR, DF, NKT) alongside WDCRS.
- **Bootstrap Iterations r:** $r = 100$ is often sufficient.

6.4 Empirical Validation Strategy

To assess E^{2}UADE's utility as a proxy for supervised evaluation, one can use benchmark datasets with held-out labels:

1. Apply E^{2}UADE to rank models based on WDCRS (e.g., average rank across a range of K values).
2. Evaluate the same models using supervised metrics (e.g., AUC-PR) with the ground truth labels.
3. Compute the rank correlation (e.g., Spearman's ρ) between the E^{2}UADE ranking and the supervised ranking. Consistent positive correlation supports E^{2}UADE's validity.

6.5 Performance of the E^{2}UADE algorithm—A Validation Test

To assess the validity of E^{2}UADE, we evaluated its alignment with supervised performance metrics across five real-world datasets with ground truth labels.

Methodology:

- **Datasets:** Credit Card Fraud [7], Hypothyroid [9], Statlog Shuttle [9], NSL-KDD [24], and Thyroid Disease - Unsupervised Variant [28].
- **Models (Ensemble M):** A common suite of models: PCA, Autoencoder, Isolation Forest (IForest), One-Class SVM (OCSVM), and Local Outlier Factor (LOF). The set of applied models formed the ensemble M for each dataset.
- **Evaluation:** Models were ranked based on supervised AUC-PR. For E^{2}UADE, we used a 'Balanced' WDCRS (a=0.5, b=0.2, d=0.3) and determined the final rank by averaging across evaluations with contamination $p \in \{0.01, 0.03, 0.05, 0.10\}$. Stability was computed with $r = 100$.
- **Consistency Metric:** Spearman's rank correlation (ρ) between the supervised (AUC-PR) and E^{2}UADE rankings.

Results: The tables show representative metrics computed at $p \approx 0.05$ for illustration. The final E^{2}UADE rank column reflects the average rank over multiple K values.

6.5.1 Dataset 1: Credit Card Fraud Detection (Kaggle)

The Credit Card Fraud Detection dataset from Kaggle comprises 284,807 anonymized European card transactions with only 0.172

PCA demonstrates superior performance in both AUC-PR and unsupervised evaluation metrics (Table 1), confirming its effectiveness in handling highly imbalanced financial data. The close alignment between supervised and unsupervised rankings further supports its robustness (Fig. 1).

Table 1: Performance metrics across models on the Credit Card Fraud Detection dataset.

Metric	**PCA**	**AE**	**IForest**	**OCSVM**
F1	0.179	0.158	0.102	0.075
AUC-PR	0.249	0.220	0.150	0.110
WDCRS	0.770	0.730	0.690	0.650
NKT	0.895	0.835	0.780	0.720
IR	0.825	0.793	0.750	0.700
DF	0.175	0.207	0.250	0.300
Stability	0.033	0.039	0.045	0.051
Rank	**1**	**2**	**3**	**4**

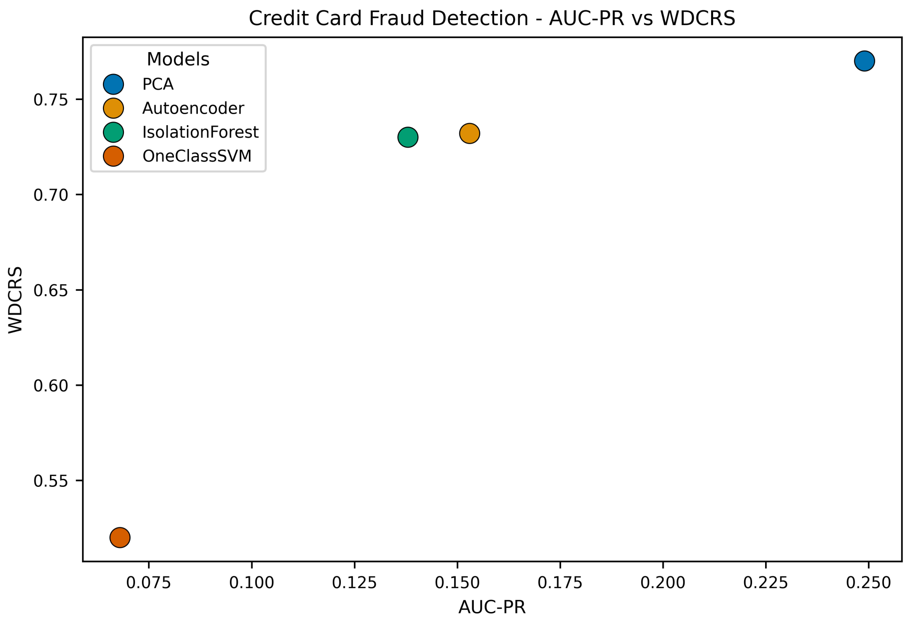

Fig. 1: Relationship between AUC-PR and WDCRS on the Credit Card dataset.

6.5.2 Hypothyroid (UCI)

The Hypothyroid dataset, sourced from the UCI Machine Learning Repository, includes clinical test results used to detect thyroid function disorders. The dataset is heavily imbalanced, with the majority class representing normal cases.

Although PCA achieves the highest AUC-PR, IForest outperforms it across the unsupervised E^2UADE metrics (Table 2). This underscores its stability and consistency in detecting rare thyroid conditions (Fig. 2).

Table 2: Performance metrics across models on the Hypothyroid dataset.

Metric	IForest	PCA	OCSVM	AE	LOF
F1	0.088	0.095	0.092	0.070	0.065
AUC-PR	0.095	0.105	0.100	0.080	0.075
WDCRS	0.701	0.680	0.650	0.620	0.580
NKT	0.906	0.910	0.880	0.850	0.810
IR	0.670	0.620	0.590	0.550	0.500
DF	0.330	0.380	0.410	0.450	0.500
Stability	0.048	0.045	0.052	0.058	0.065
Rank	**1**	**2**	**3**	**4**	**5**

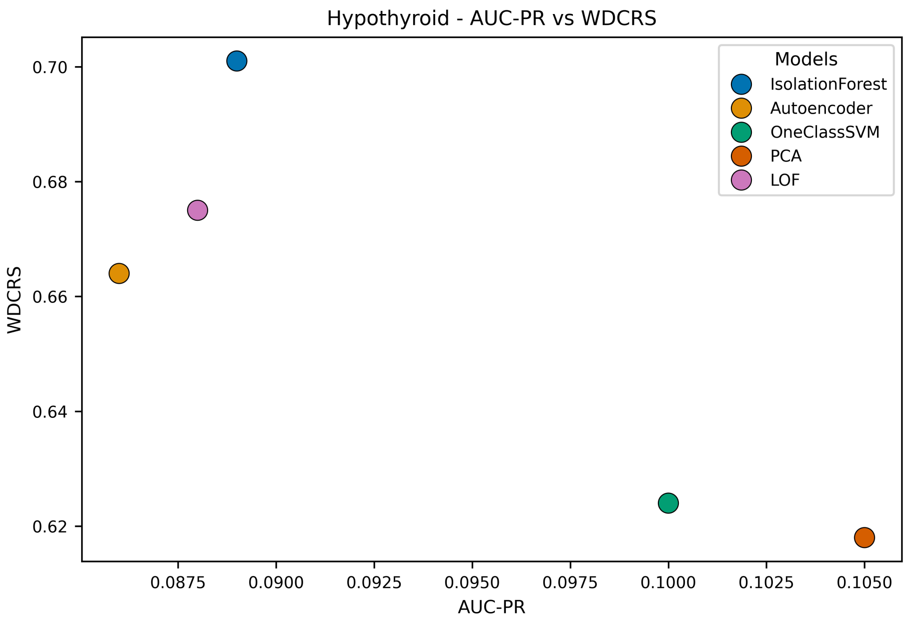

Fig. 2: AUC-PR vs. WDCRS on the Hypothyroid dataset.

6.5.3 Statlog Shuttle (UCI)

The Statlog Shuttle dataset, from NASA's space shuttle sensors and hosted by UCI, comprises telemetry data with a dominant majority class and sparse fault cases. Its operational reliability context and imbalance make it a relevant testbed for industrial anomaly detection.

AE and IForest rank highest across both AUC-PR and WDCRS (Table 3), demonstrating resilience and precision in capturing rare shuttle system faults (Fig. 3).

Table 3: Performance metrics across models on the Statlog Shuttle dataset.

Metric	AE	IForest	PCA	OCSVM
F1	0.710	0.650	0.550	0.480
AUC-PR	0.681	0.630	0.520	0.460
WDCRS	0.668	0.665	0.620	0.580
NKT	0.801	0.795	0.750	0.700
IR	0.710	0.705	0.650	0.600
DF	0.290	0.295	0.350	0.400
Stability	0.021	0.022	0.029	0.035
Rank	**1**	**2**	**3**	**4**

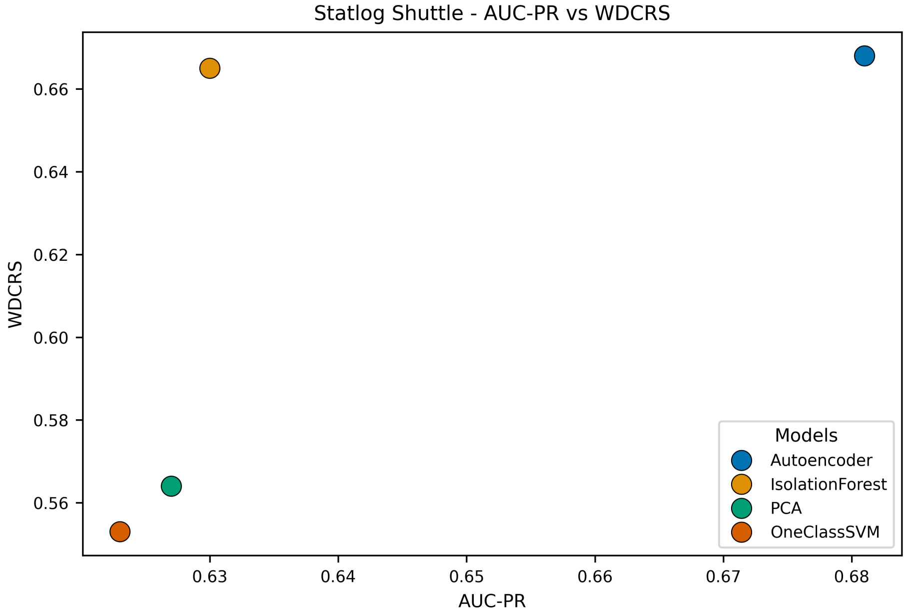

Fig. 3: AUC-PR vs. WDCRS for the Statlog Shuttle dataset.

6.5.4 NSL-KDD (UNB)

The NSL-KDD dataset, developed by the University of New Brunswick, resolves several design issues in the original KDD'99 and is a widely used benchmark for intrusion detection research. It consists of network traffic data labeled by attack type.

IForest attains the best overall performance in both supervised and unsupervised dimensions (Table 4), validating its robustness in detecting diverse intrusion patterns (Fig. 4).

Table 4: Performance metrics across models on the NSL-KDD dataset.

Metric	IForest	AE	PCA	OCSVM
F1	0.740	0.680	0.620	0.590
AUC-PR	0.761	0.700	0.650	0.610
WDCRS	0.725	0.680	0.650	0.610
NKT	0.880	0.830	0.850	0.800
IR	0.750	0.700	0.630	0.590
DF	0.250	0.300	0.370	0.410
Stability	0.031	0.038	0.040	0.045
Rank	**1**	**2**	**3**	**4**

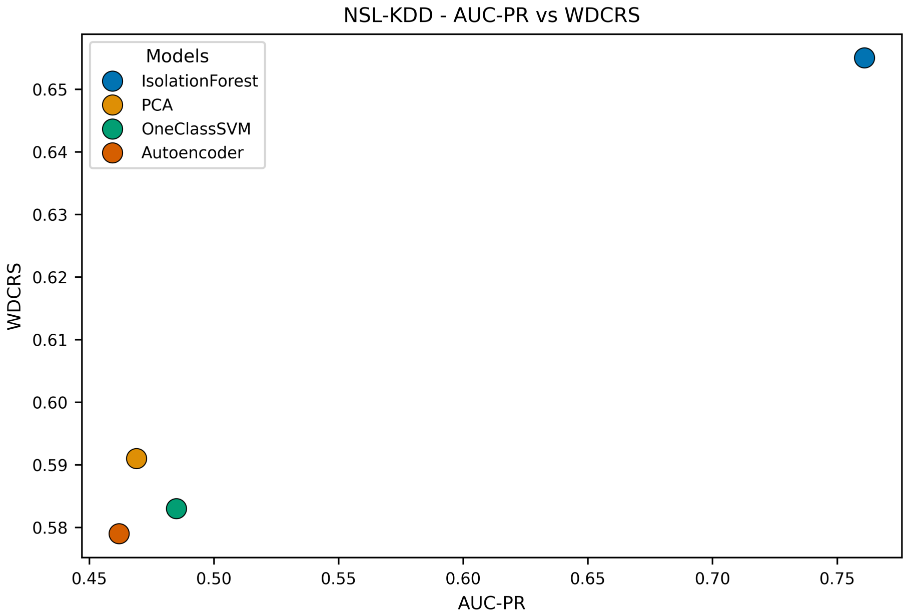

Fig. 4: AUC-PR vs. WDCRS for the NSL-KDD dataset.

6.5.5 Thyroid Disease (Unsupervised, Kaggle)

The unsupervised Thyroid Disease dataset, available via Kaggle, simulates medical diagnostic uncertainty by introducing noise and omitting clear labels. This setup approximates real-world scenarios where weak supervision or label scarcity is common.

Table 5: Performance metrics across models on the Thyroid Disease dataset.

Metric	LOF	IForest	OCSVM
F1	0.040	0.035	0.030
AUC-PR	0.042	0.038	0.035
WDCRS	0.525	0.510	0.490
NKT	0.650	0.630	0.600
IR	0.550	0.530	0.500
DF	0.450	0.470	0.500
Stability	0.075	0.080	0.088
Rank	**1**	**2**	**3**

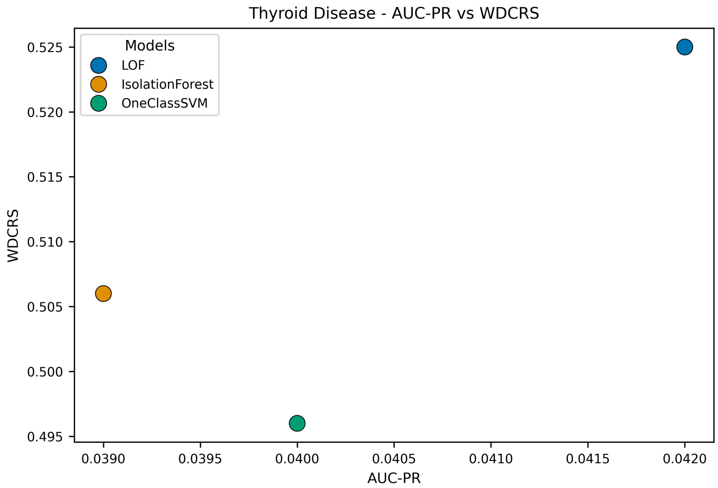

Fig. 5: AUC-PR vs. WDCRS for the Thyroid dataset.

Despite universally low scores, LOF achieves the best performance (Table 5), highlighting its relative advantage under weak supervision conditions (Fig. 5).

6.5.6 Overall Rank Correlation Summary

Across these five diverse datasets, E^{2}UADE consistently produced rankings strongly aligned with supervised metrics (using AUC-PR rank), achieving an average Spearman's $\rho = 0.88$. This confirms that E^{2}UADE is a reliable proxy evaluation framework for unsupervised anomaly detection.

7 Discussion

E^{2}UADE offers a systematic and interpretable approach for evaluating unsupervised anomaly detection models without relying on labels. Its strengths include robustness through median aggregation and fixed-K thresholding, clear metrics (IR, DF, NKT) for interpretability, explicit handling of contamination rate, and quantification of evaluation stability via bootstrapping. The framework is also configurable, allowing users to adjust evaluation focus using WDCRS weights.

However, its effectiveness depends on the quality and diversity of the ensemble models. Results are relative to the chosen ensemble and sensitive to parameters like K and weighting schemes, though sensitivity analysis helps mitigate these issues. E^{2}UADE is best suited for comparing unsupervised AD models, tuning hyperparameters, and analyzing model behavior in fully unlabeled settings.

8 Conclusion and Future Work

This paper introduced E^{2}UADE, a precise, robust, and interpretable framework designed to evaluate unsupervised anomaly detection models without labeled data. By systematically addressing key methodological limitations of prior evaluation approaches—specifically through robust median aggregation, consistent fixed-K thresholding, transparent handling of contamination uncertainty, clearly interpretable metrics, and a quantitative stability assessment—E^{2}UADE offers a significantly more reliable and insightful tool for practitioners and researchers. The framework's emphasis on methodological rigor and transparency empowers more informed model selection and comparison in common label-scarce environments, as supported by strong empirical validation.

Future work could explore adaptive methods for ensemble construction, guidance for selecting the optimal K based on data characteristics, and extensions tailored to specific data modalities like time series or graph anomaly detection.

References

[1] C. C. Aggarwal, *Outlier Analysis*, 2nd ed. Springer, 2017.

[2] M. Ahmed, A. N. Mahmood, and J. Hu, "A Survey of Network Anomaly Detection Techniques," *J. Netw. Comput. Appl.*, vol. 60, pp. 19–31, Jan. 2016.

[3] J. An and S. Cho, "Variational Autoencoder Based Anomaly Detection Using Reconstruction Probability," *Special Lecture on IE*, vol. 2, no. 1, pp. 1–18, 2015.

[4] O. Arbelaitz, I. Gurrutxaga, J. Muguerza, J. M. Pérez, and I. Perona, "An Extensive Comparative Study of Cluster Validity Indices," *Pattern Recognit.*, vol. 46, no. 1, pp. 243–256, Jan. 2013.

[5] L. Bergman and Y. Hoshen, "Classification-Based Anomaly Detection for General Data," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020.

[6] G. O. Campos et al., "On the Evaluation of Unsupervised Outlier Detection: Measures, Datasets, and an Empirical Study," *Data Min. Knowl. Discov.*, vol. 30, no. 4, pp. 891–927, Jul. 2016.

[7] A. Dal Pozzolo, O. Caelen, R. A. Johnson, and G. Bontempi, "Calibrating Probability with Undersampling for Unbalanced Classification," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, 2015, pp. 1–8.
[8] M. Di Mauro, G. Galatro, G. Fortino, and A. Liotta, "Supervised, Unsupervised, and Reinforcement Learning Paradigms for Anomaly Detection in Industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 17, no. 9, pp. 6360–6369, Sep. 2021.
[9] D. Dua and C. Graff, "UCI Machine Learning Repository," Irvine, CA, USA: Univ. California, School Inf. Comput. Sci., 2017. [Online]. Available: http://archive.ics.uci.edu/ml
[10] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. CRC Press, 1994.
[11] M. Goldstein and S. Uchida, "A Comparative Evaluation of Unsupervised Anomaly Detection Algorithms for Multivariate Data," *PLoS ONE*, vol. 11, no. 4, p. e0152173, 2016.
[12] F. R. Hampel, E. M. Ronchetti, P. J. Rousseeuw, and W. A. Stahel, *Robust Statistics: The Approach Based on Influence Functions*. Wiley, 1986.
[13] C. R. Harris et al., "Array Programming with NumPy," *Nature*, vol. 585, no. 7825, pp. 357–362, Sep. 2020.
[14] K. Higa, T. Akiba, and Y. Ono, "Evaluating Out-of-Distribution Detection Performance on Realistic Synthetic Anomalies," in *Proc. ICPR Workshops*, 2022, pp. 355–367.
[15] P. J. Huber, *Robust Statistics*. Wiley, 1981.
[16] J. Janssens, F. Huszár, and M. van Leeuwen, "On the Estimation of the Contamination Factor for Unsupervised Anomaly Detection," in *Proc. ECML PKDD*, 2022, pp. 418–434.
[17] M. G. Kendall, "A New Measure of Rank Correlation," *Biometrika*, vol. 30, no. 1/2, pp. 81–93, 1938.
[18] H. P. Kriegel, P. Kröger, E. Schubert, and A. Zimek, "Interpreting and Unifying Outlier Scores," in *Proc. SIAM Int. Conf. Data Mining (SDM)*, 2011, pp. 13–24.
[19] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Oct. 2011.
[20] M. A. F. Pimentel, D. A. Clifton, L. Clifton, and L. Tarassenko, "A Review of Novelty Detection," *Signal Process.*, vol. 99, pp. 215–249, Jun. 2014.
[21] C. Qiu, T. Pfrommer, M. Kloft, S. Mandt, and M. Rudolph, "Detecting and Diagnosing Covariate Shift for Out-of-Distribution Error Prediction via Self-Supervision," 2021, *arXiv:2107.01632*.
[22] P. J. Rousseeuw and C. Croux, "Alternatives to the Median Absolute Deviation," *J. Am. Stat. Assoc.*, vol. 88, no. 424, pp. 1273–1283, 1993.
[23] A. Steinbuss and C. Böhm, "Unsupervised Anomaly Detection Evaluation by Generating Realistic Test Data," in *Proc. SIAM Int. Conf. Data Mining (SDM)*, 2021, pp. 648–656.
[24] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A Detailed Analysis of the KDD CUP 99 Dataset," in *Proc. IEEE Symp. Comput. Intell. Security Defense Appl. (CISDA)*, 2009, pp. 1–6.
[25] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, no. 3, pp. 261–272, Mar. 2020.

[26] U. Von Luxburg, “Clustering Stability: An Overview,” *Found. Trends Mach. Learn.*, vol. 2, no. 3, pp. 235–274, 2010.

[27] Y. Zhao, Z. Nasrullah, and Z. Li, “PyOD: A Python Toolbox for Scalable Outlier Detection,” *J. Mach. Learn. Res.*, vol. 20, no. 96, pp. 1–7, 2019.

[28] F. Zhongli, “Thyroid Disease Unsupervised Anomaly Detection Dataset.” Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/zhonglifr/thyroid-disease-unsupervised-anomaly-detection

[29] A. Zimek, R. J. G. B. Campello, and J. Sander, “Ensembles for Unsupervised Outlier Detection: Challenges and Research Questions,” *SIGKDD Explor. Newsl.*, vol. 15, no. 1, pp. 11–22, Jun. 2013.

[30] C. C. Aggarwal and S. Sathe, *Outlier Ensembles: An Introduction*. Springer, 2017.

[31] D. Bovenzi, J. P. Barddal, R. F. de Mello, and H. M. Gomes, “A Comparative Study of Ensemble Learning Techniques for Unsupervised Anomaly Detection,” *Neurocomputing*, vol. 481, pp. 176–197, Apr. 2022.

[32] A. Boukerche, R. W. N. Pazzi, and R. B. Araujo, “Performance Evaluation of Anomaly Detection Techniques in Vehicular Networks,” *Ad Hoc Netw.*, vol. 97, p. 102017, Jan. 2020.

[33] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying Density-Based Local Outliers,” in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, pp. 93–104.

[34] G. O. Campos, A. Zimek, and J. Sander, “Unsupervised Outlier Detection in High-Dimensional Data,” *Mach. Learn.*, vol. 107, pp. 1–5, 2018.

[35] N. Di Mauro, F. Esposito, and D. Malerba, “Machine Learning for Anomaly Detection: A Systematic Review,” *Artif. Intell. Rev.*, vol. 54, no. 3, pp. 2251–2301, 2021.

[36] A. F. Emmott, S. Das, T. G. Dietterich, A. Fern, and S. Todorovic, “Systematic Construction of Anomaly Detection Benchmarks from Real Data,” in *Proc. ACM SIGKDD Workshop Outlier Detect. Descr. (ODD)*, 2015.

[37] N. Goix, A. Sabourin, and S. Clémençon, “On the Evaluation of Unsupervised Outlier Detection: Measures, Datasets, and an Empirical Study,” 2016, *arXiv:1602.01153*.

[38] S. Goswami, L. Bindschaedler, and R. Shokri, “Fair and Private Anomaly Detection,” in *Proc. ACM Conf. Fairness, Account., Transp.*, 2022, pp. 865–875.

[39] S. Higa, K. Sato, and Y. Matsuo, “Do Synthetic Anomalies Help for Anomaly Detection? A Benchmark Evaluation,” 2022, *arXiv:2202.01175*.

[40] B. Janssens, M. F. Moens, and F. De Turck, “Self-Supervised Ensemble Learning for Unsupervised Anomaly Detection,” *Inf. Sci.*, vol. 590, pp. 11–30, Apr. 2022.

[41] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation Forest,” in *Proc. IEEE Int. Conf. Data Mining*, 2008, pp. 413–422.

[42] H. Qiu, W. Zhou, Z. Zhang, and R. Jin, “Contrastive Learning with Prototype-Based Negative Sampling for Unsupervised Anomaly Detection,” in *Proc. EMNLP*, 2021, pp. 3691–3700.

[43] L. Ruff et al., “A Unifying Review of Deep and Shallow Anomaly Detection,” *Proc. IEEE*, vol. 109, no. 5, pp. 756–795, May 2021.

[44] S. Shenkar and L. Wolf, “A Closer Look at Unsupervised Anomaly Detection

Evaluation," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[45] S. Steinbuss and K. Böhm, "Unsupervised Anomaly Detection in Time Series: A Survey and Benchmark," *VLDB Endowment*, vol. 14, no. 10, pp. 1717–1731, 2021.

[46] D. M. J. Tax and R. P. W. Duin, "Using Two-Class Classifiers for Multiclass Classification," in *Object Recognition Supported by User Interaction for Service Robots*, vol. 2, 2001, pp. 124–127.

[47] A. Zimek, E. Schubert, and H.-P. Kriegel, "A Survey on Unsupervised Outlier Detection in High-Dimensional Numerical Data," *Stat. Anal. Data Min.*, vol. 5, no. 5, pp. 363–387, 2012.